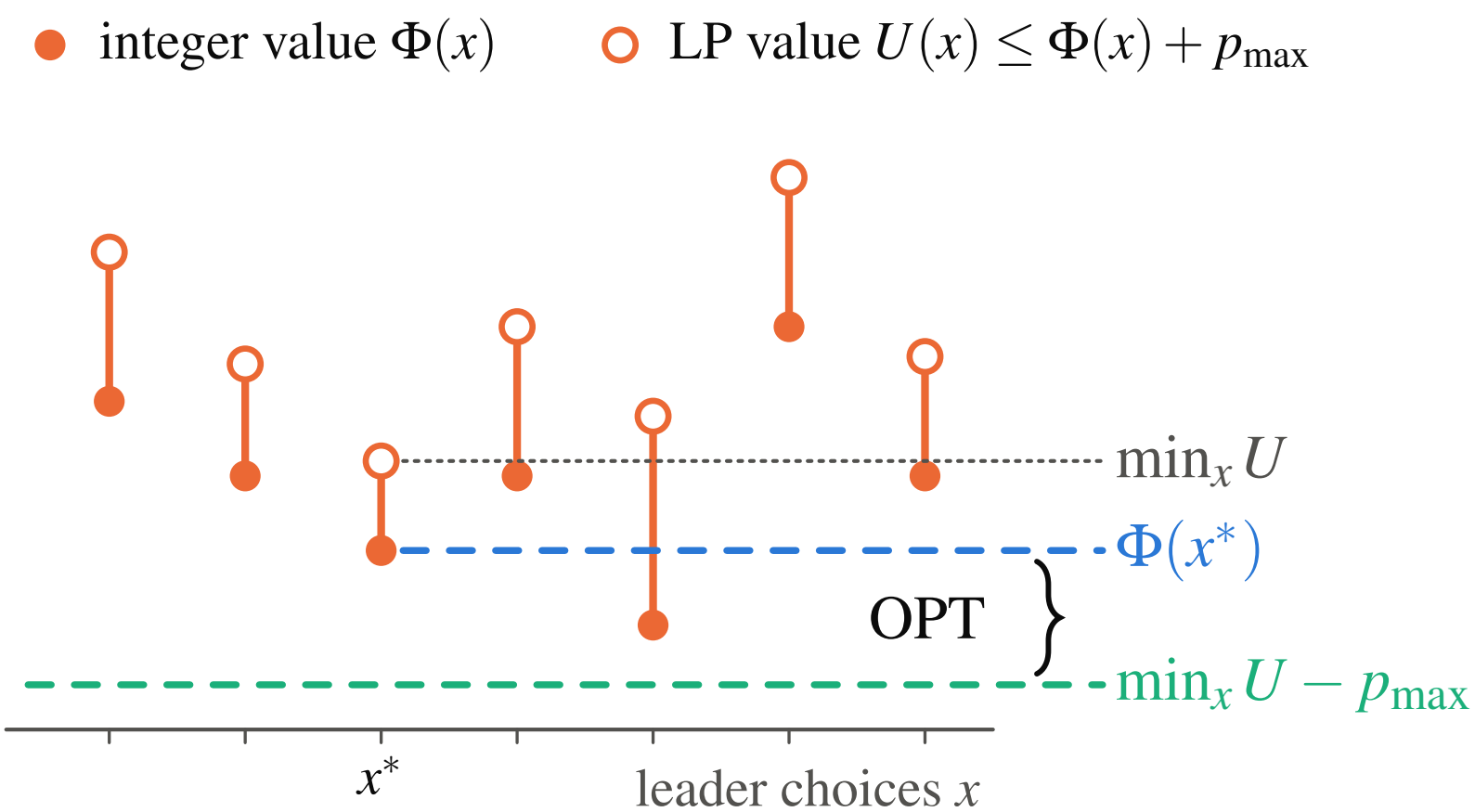
integer value $\Phi(x)$
LP value $U(x) \leq \Phi(x) + p_{\max}$
$\min_x U$
$\Phi(x^*)$
OPT
$\min_x U - p_{\max}$
$x^*$
leader choices $x$